Network Configuration Report
School Infrastructure

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Network Configuration Document

Note: The main server (DHCP Server on Core Switch 1) is located in Building A (Building 1).

Core Switch 1 (Building 1) Configuration

VLANs and IP Configuration

```
configure terminal
vlan 10
name Lab_A
exit
vlan 20
name Lab_B
exit
vlan 30
name Managers
exit
vlan 40
name wifi
exit

interface vlan 1
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

DHCP Configuration

```
ip dhcp pool VLAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

```
exit

ip dhcp pool VLAN20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit

ip dhcp pool VLAN30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
exit

ip dhcp pool VLAN40
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 8.8.8.8
exit

ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.40.1
```

Trunk Link to Core Switch 2

```
interface GigabitEthernet1/1/1
switchport mode trunk
switchport trunk allowed vlan 1,10,20,30,40
exit
```

Ports for Access Points (AP1 to AP8)

```
interface range GigabitEthernet1/0/6 - 18
switchport mode access
switchport access vlan 40
exit
```

Port to Router (4331)

```
interface GigabitEthernet1/0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

Static Route to Router

```
ip route 0.0.0.0 0.0.0.0 192.168.2.2
```

Core Switch 2 (Building 2) Configuration

VLANs and IP Configuration

```
configure terminal
vlan 10
name Lab_A
exit
vlan 20
name Lab_B
exit
vlan 40
name wifi
exit

interface vlan 1
ip address 192.168.1.2 255.255.255.0
no shutdown
exit

interface vlan 10
ip helper-address 192.168.1.1
exit

interface vlan 20
ip helper-address 192.168.1.1
exit

interface vlan 40
ip helper-address 192.168.1.1
exit
```

Trunk Link to Core Switch 1

```
interface GigabitEthernet1/1/1
switchport mode trunk
switchport trunk allowed vlan 1,10,20,40
exit
```

Ports for Access Points (AP9 to AP16)

```
interface range GigabitEthernet0/1 - 8
switchport mode access
switchport access vlan 40
exit
```

Access Switch Lab A (Building 1) Configuration

VLANs and Ports

```
configure terminal
vlan 10
name Lab_A
exit
```

```
vlan 40
name wifi
exit

interface range FastEthernet0/1 - 24
switchport mode access
switchport access vlan 10
exit

interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,40
exit
```

Access Switch Lab B (Building 1) Configuration

VLANs and Ports

```
configure terminal
vlan 20
name Lab_B
exit

interface range FastEthernet0/1 - 24
switchport mode access
switchport access vlan 20
exit

interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 20
exit
```

Access Switch 3 (Building 2) Configuration

VLANs and Ports

```
configure terminal
vlan 10
name Lab_A
exit
vlan 40
name wifi
exit

interface range FastEthernet0/1 - 24
switchport mode access
switchport access vlan 10
exit

interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,40
exit
```

Access Switch Lab B (Building 2) Configuration

VLANs and Ports

```
configure terminal
vlan 20
name Lab_B
exit

interface range FastEthernet0/1 - 24
switchport mode access
switchport access vlan 20
exit

interface GigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 20
exit
```

Router (Cisco ISR 4331) Configuration

Interface Configuration

```
configure terminal
interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.0
no shutdown
exit

interface GigabitEthernet0/0/0
ip address 203.0.113.2 255.255.255.0
no shutdown
exit
```

Static Route

```
ip route 192.168.1.0 255.255.255.0 192.168.2.1
ip route 192.168.10.0 255.255.255.0 192.168.2.1
ip route 192.168.20.0 255.255.255.0 192.168.2.1
ip route 192.168.30.0 255.255.255.0 192.168.2.1
ip route 192.168.40.0 255.255.255.0 192.168.2.1
```

NAT Configuration for Internet Access

```
access-list 1 permit 192.168.1.0 0.0.0.255
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255

ip nat inside source list 1 interface GigabitEthernet0/0/0 overload
```

```
interface GigabitEthernet0/0/1
ip nat inside
exit

interface GigabitEthernet0/0/0
ip nat outside
exit
```

Cloud-PT (Internet) Configuration

Interface Configuration (Done via GUI in Packet Tracer)

- Interface: Ethernet6 (Connected to Router's GigabitEthernet0/0/0)

- IP Address: 203.0.113.1

- Subnet Mask: 255.255.255.0

Access Points (AP1 to AP16) Configuration

Wireless Configuration (Done via GUI in Packet Tracer)

- SSID: We

- Passphrase: 12345678

- IP Configuration: DHCP (Obtains IP from VLAN 40, e.g., 192.168.40.x)

Password Configuration for All Devices (Core Switches, Access Switches, Router, Firewall)

Core Switch 1

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Core Switch 2

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Access Switch 1

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Access Switch 2

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Access Switch 3

```
configure terminal
line console 0
password 123
login
```

```
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Access Switch 4

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Router (4331)

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
login
exit

enable password 123
service password-encryption
exit
```

Firewall (5506)

```
configure terminal
line console 0
password 123
login
exit

line vty 0 4
password 123
```

```
login
exit

enable password 123
service password-encryption
exit
```

Network Features and Information

Network Overview

This network is designed to serve two buildings (Building A and Building B) in a school environment. The school has a total of 98 devices connected to the network, including PCs, laptops, and wireless devices. The network uses a hierarchical design with Core Switches, Access Switches, Access Points, a Router, and a Firewall to ensure efficient traffic management, security, and scalability. The network is segmented into VLANs for better organization and security:

- VLAN 10 (Lab_A): For lab devices in Building A and B.

- VLAN 20 (Lab_B): For other lab devices in Building A and B.

- VLAN 30 (Managers): For management devices.

- VLAN 40 (wifi): For wireless devices connected via Access Points.

The DHCP Server on Core Switch 1 (Building A) dynamically assigns IPs to all devices across VLANs, ensuring centralized IP management. Access Points provide wireless connectivity with the SSID `We` and passphrase `12345678`, allowing devices to connect seamlessly and obtain IPs automatically.

To protect the network from viruses and attacks, a free antivirus software, **Avast Free Antivirus**, will be installed on all 98 devices. Avast Free Antivirus provides real-time protection against malware, ransomware, and phishing attacks, ensuring the safety of the school's devices and data.

Devices Used, Speeds, and Reasons for Selection

Core Switches: Cisco Catalyst 3650

- Speed : Supports up to 10 Gbps with fiber optic uplinks and 1 Gbps for copper ports.

- Reasons for Use : Used as the backbone of the network to handle high traffic between buildings and VLANs, perform Inter-VLAN routing, and host the DHCP Server.

- Advantages :

- Supports fiber optic modules for high-speed uplinks, making it ideal for connecting buildings.

- High performance with Layer 3 capabilities for routing.

- Cost-effective for small to medium-sized networks.

- Scalable with support for stacking multiple switches.

Access Switches: Cisco Catalyst 2960

- Speed: Supports up to 1 Gbps for copper ports and 1 Gbps uplinks.

- Reasons for Use: Used to connect end devices (like PCs and Access Points) in each building, providing reliable access-layer connectivity.

- Advantages:

- Affordable and reliable for access-layer switching.

- Supports VLANs and basic security features.

- Energy-efficient with Cisco EnergyWise technology.

Access Points: AP-PT-AC (Packet Tracer Simulated Access Points)

- Speed: Supports up to 300 Mbps (simulated Wi-Fi speed in Packet Tracer).

- **Reasons for Use**: Provides wireless connectivity for classrooms in both buildings, allowing students and staff to connect via Wi-Fi.

- Advantages:

- Easy to configure with a simple SSID (`We`) and passphrase (`12345678`).

- Supports VLAN tagging (VLAN 40) for wireless traffic.

- Simulates modern Wi-Fi standards like 802.11ac in Packet Tracer.

Router: Cisco ISR 4331

- Speed: Supports up to 1 Gbps throughput (depending on configuration).

- Reasons for Use: Used to connect the internal network to an external network (e.g., the Internet) if needed.

- Advantages:

- High performance with support for advanced routing protocols.

- Modular design allows for adding interfaces (e.g., fiber, LTE).

- Built-in security features like VPN and firewall capabilities.

Firewall: Cisco ASA 5506

- Speed: Supports up to 750 Mbps firewall throughput.

- Reasons for Use: Used to secure the network by filtering traffic between the internal network and external networks.

- Advantages:

- Provides robust security with stateful inspection and VPN support.

- Compact design, suitable for small to medium networks.

- Supports advanced threat detection and prevention.

Network Advantages

- Scalability: The use of VLANs (Lab_A, Lab_B, Managers, wifi) and Core Switches allows the network to scale easily by adding more devices or VLANs.

- Centralized Management: The DHCP Server on Core Switch 1 simplifies IP management across both buildings.

- Security: VLAN segmentation isolates traffic (e.g., Lab_A devices can't access Managers' devices), passwords (`123`) on all devices prevent unauthorized access, and Avast Free Antivirus protects all 98 devices from viruses and attacks.

- Reliability: Cisco devices (3650, 2960, 4331, 5506) are known for their reliability and performance.

- Wireless Coverage: 16 Access Points (8 per building) ensure comprehensive Wi-Fi coverage for classrooms, with a unified SSID (`We`) for seamless connectivity.

- Internet Access: The router (Cisco ISR 4331) is configured with NAT to allow all internal devices to access the Internet via a Cloud-PT device, without affecting internal network functionality.